

A Lean-Checked Finite Farey Mediator Theorem

Source Provenance, an Obstructed Endpoint Taxonomy, and a Direct Euclidean Constructor

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Abstract

A mathematical proof route can fail even when its target theorem remains true. This paper presents a finite Farey example in which an attempted classification of endpoint pairs is refuted by a concrete counterexample, while a separate Euclidean constructor proves the intended mediator target. The carrier is the finite list of reduced fractions between zero and one whose denominators do not exceed a fixed bound N . The target states that two increasing carrier elements with determinant greater than one admit another listed carrier element strictly between them.

The formal development records two complementary results. First, the pair $0/1 < 2/3$ at $N = 3$ lies outside a proposed endpoint-shape union, refuting that universal taxonomy. Second, a bounded modular-inverse construction, completed by Euclidean descent, supplies the direct finite mediator route. The terminal Lean theorems establish the bounded inverse contract, the certified mediator constructor premise, and the listed-mediator target in the stated formalization.

The release package pins Lean 4.30.0, the relevant dependency manifest, source hashes, theorem names, axiom reports, and exact replay instructions. A corrected counterexample source replaces four historical `native_decide` calls with kernel reduction; this removes the generated native-decision axioms from the promoted obstruction proof. An independent enumerator additionally checks 1,056,006 determinant-greater-than-one endpoint pairs through $N = 40$, finding a listed strict mediator for every one. The proved result is a finite combinatorial theorem and a reproducible proof-route case study. Its role in the broader Super Information Theory program is to establish a disciplined finite formal component from which any later analytic or physical lift must proceed explicitly.

modular inverse; constructive proof route; theorem provenance; Super Information Theory

Keywords: Farey sequence; formal verification; Lean 4; Euclidean algorithm;

Scope and evidence boundary

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1. The Problem Before the Terminology

Suppose a program must establish that an intermediate state exists between two allowed endpoints. One strategy is to sort every endpoint pair into a small collection of shapes and provide a witness for each shape. Another strategy is to build the witness directly from the arithmetic data of the endpoints.

These strategies are logically different. A counterexample to the first strategy shows that its classification is incomplete. It does not show that the intermediate state is absent. If a direct construction succeeds, the target survives even though the taxonomy fails.

This distinction matters in formal proof development because internal route labels can easily be mistaken for theorem statements. A failed helper lemma can be reported as if the target had failed, or a successful

finite helper can be promoted beyond the theorem it actually proves. The solution is to expose five objects separately:

1. the finite carrier;
2. the order and determinant on that carrier;
3. the attempted endpoint taxonomy;
4. the direct witness constructor; and
5. the exact terminal theorem.

Only after these operations are clear do the internal artifact names BC-054IS, BC-055M, and BC-055O become useful. They are labels for stages of the proof route, not substitutes for the mathematics.

2. Established Farey Structure

For a positive integer N , define the finite carrier

$$\mathcal{F}_N = \left\{ \frac{p}{q} \in \mathbb{Q} \mid 0 \leq p \leq q \leq N, \quad q \geq 1, \quad \gcd(p, q) = 1 \right\}. \quad (1)$$

The formal source represents each fraction as an ordered pair (p, q) . For two reduced pairs $a = (p, q)$ and $b = (r, s)$, strict order is tested by cross multiplication:

$$a < b \iff ps < rq. \quad (2)$$

The oriented determinant is

$$\Delta(a, b) = rq - ps. \quad (3)$$

This is the familiar determinant that distinguishes neighboring and non-neighboring Farey fractions. Consecutive elements of a Farey sequence have determinant one. The formal target studied here concerns the complementary case: if two listed fractions are in increasing order and their determinant is greater than one, the finite list should contain another fraction between them.

Written independently of the implementation, the target is

$$\begin{aligned} \forall N, \forall a, b \in \mathcal{F}_N, \quad & a < b \wedge \Delta(a, b) > 1 \\ \implies & \exists c \in \mathcal{F}_N : a < c < b. \end{aligned} \quad (4)$$

Equation (4) is what the Lean source calls the arbitrary endpoint listed-mediator target. The terminology is earned by the operation: *listed* means that the witness occurs in the explicitly generated finite carrier, and *mediator* means that it is strictly between the endpoints.

3. The Classification Route and Its Counterexample

An earlier proof route attempted to cover every endpoint pair satisfying the antecedent of Eq. (4) with a union of several parametrized endpoint shapes. Each admitted shape had its own witness construction. If the union had covered every relevant pair, the shape-specific lemmas would have implied the target.

The universal coverage claim was false. At $N = 3$, choose

$$\begin{aligned} a &= \frac{0}{1}, \\ b &= \frac{2}{3}. \end{aligned} \tag{5}$$

Both fractions belong to $\mathcal{F}_3$, they are strictly ordered, and

$$\Delta(a, b) = 2 \cdot 1 - 0 \cdot 3 = 2 > 1. \tag{6}$$

The formal proof establishes that this pair is not contained in the proposed shape union. Therefore the universal endpoint-shape bridge is refuted as stated.

This obstruction diagnoses the route rather than the target. Indeed, the same finite carrier already contains

$$\frac{0}{1} < \frac{1}{3} < \frac{2}{3} \quad \text{and} \quad \frac{0}{1} < \frac{1}{2} < \frac{2}{3}. \tag{7}$$

The counterexample therefore carries two pieces of information at once:

- the proposed classification is incomplete; and
- the listed-mediator conclusion remains true for the counterexample.

The corrected artifact BC-054IS records exactly this distinction. Its counterexample theorem is a permanent negative result about one helper taxonomy, not a negative result about Eq. (4).

4. Direct Construction Through a Bounded Inverse

The surviving route does not classify endpoint shapes. It extracts arithmetic data directly.

The central bounded-inverse statement is:

$$\gcd(p, q) = 1, \quad 1 < q, \quad p < q \implies \exists x, u \in \mathbb{N} : 1 \leq x \leq q, \quad px + 1 = uq. \tag{8}$$

Equation (8) is a bounded representative of the modular inverse of $-p$ modulo q . The BC-055O source proves it by induction following the Euclidean descent of the greatest-common-divisor computation. The proof treats the $p = 1$ case directly and transports a witness from the remainder step for $1 < p < q$.

Only one endpoint orientation must be proved independently. If the left bounded-inverse result is applied to the complementary numerator $s - r$, then

$$(s - r)x + 1 = us \tag{9}$$

rearranges to

$$rx = (x - u)s + 1. \tag{10}$$

Equation (10) supplies the opposite orientation. This complement transformation is the mathematical content of the stage historically labeled BC-055M. It is included in the replayed BC-055O source, so the final result does not depend on an external prose assertion that the reduction exists.

The formal development then combines the two bounded inverse orientations with the endpoint data. It produces a pair c that satisfies:

$$\begin{aligned} c &\in \mathcal{F}_N, \\ a &< c, \\ c &< b. \end{aligned} \tag{11}$$

The implementation separates construction from list membership. First it builds a certified bounded reduced mediator. Then it applies the proved equivalence between bounded-reduced status and membership in the generated canonical list. This yields Eq. (4).

5. The Formal Theorem Chain

The terminal source `bc055o_left_endpoint_euclidean_algebra_step_20260616.lean` contains the entire dependency chain needed for the final finite target. The promoted theorems are:

Mathematical role	Lean theorem
left bounded inverse	<code>left_endpoint_bounded_inverse_contract_proved</code>
direct certified constructor	<code>direct_arbitrary_certified_mediator_constructor_premise_proved</code>
listed strict mediator	<code>arbitrary_endpoint_listed_mediator_target_contract_proved</code>

The route can be summarized as

$$\begin{aligned} \text{Euclidean descent} &\implies \text{left bounded inverse} \implies \text{right bounded inverse} \\ &\implies \text{certified mediator} \implies \text{listed mediator}. \end{aligned} \tag{12}$$

The failed and successful paths therefore form a fork:

$$\begin{aligned} \text{endpoint-shape taxonomy} &\not\Rightarrow \text{universal shape bridge}, \\ \text{bounded inverse constructor} &\implies \text{listed-mediator target}. \end{aligned} \tag{13}$$

This is the paper’s main proof-engineering result. A route-local obstruction can coexist with a theorem proved by an independent route.

6. Trust Boundary and Reproducible Replay

6.1 Correcting the historical counterexample proof

The historical BC-054IS source used `native_decide` for four small closed arithmetic goals inside the canonical-counterexample theorem. Lean accepts such proofs, but it records a bespoke axiom for each native computation. That extends the trusted computing base to the compiler and native runtime.

For the final-preprint package, the four calls were replaced by `decide`. Because the goals are small and closed, kernel reduction completes them directly. The mathematical statement and all endpoint definitions are unchanged.

The corrected axiom report is:

Corrected BC-054IS theorem	Reported dependencies
bridge_counterexample_canonical	propext, Quot.sound
bridge_counterexample_not_shape	propext, Quot.sound
arbitrary_endpoint_shape_bridge_refuted	propext, Quot.sound
explicit obstruction decision	no axioms

The generated native-decision axioms present in the historical file no longer appear in the promoted counterexample proof.

6.2 Direct-constructor report

The three promoted BC-055O theorems report Lean’s standard `propext`, `Classical.choice`, and `Quot.sound` dependencies. The explicit closure decision reports no axioms. The source contains no locally declared axiom and no `sorry`, `admit`, or `unsafe` placeholder.

The distinction is important. “Lean-checked” here means that the exact theorem types elaborate and close under the pinned environment and reported logical dependencies. It does not mean that an informal physical interpretation has been imported into the proof.

6.3 Pinned environment

The release package records:

Item	Pinned identity
Lean	4.30.0
Lean commit	d024af099ca4bf2c86f649261ebf59565dc8c622
Lake	5.0.0-src+d024af0
Mathlib	c5ea00351c28e24afc9f0f84379aa41082b1188f
counterexample source	corrected kernel-reducible BC-054IS copy
constructor source	exact BC-055O copy

Both promoted source files were checked through the repository’s serialized Lean compiler lease. Exact source hashes and replay receipts accompany the PDF.

7. Independent Finite Census

A separate Python enumerator mirrors Eqs. (1)-(4) without importing the Lean implementation. For every N from 1 through 40, it:

1. generates every reduced fraction p/q with $0 \leq p \leq q \leq N$;
2. orders the fractions exactly using rational arithmetic;
3. enumerates every increasing endpoint pair;
4. selects every pair with determinant greater than one; and
5. checks for a listed fraction strictly between the endpoints.

The census produced:

Quantity	Result
ordered endpoint pairs	1,069,924
determinant-greater-than-one targets	1,056,006
targets with a listed strict mediator	1,056,006
failures	0

This is a cross-implementation consistency check. It is not needed to make the universal Lean result true, but it helps expose definition mismatches that can otherwise survive inside a single formal system.

8. Provenance of the Proof Route

Version 1 was published under the historical title *Jackie Scott SIT Kernel Anchor*. It was a source-governance paper rather than a theorem paper. It recovered several heterogeneous research leads:

- private discussions that pointed toward SIT action, current, operator, and falsifier questions;
- comparison material associated with Jackie Blankenship and Scott Cave;
- Super Information Theory source drafts;
- a proof-shape analogy drawn from a unit-distance theorem; and
- candidate experimental routes involving condensates, clocks, and resonance.

Those materials played different roles. Some were provenance leads, some were comparison context, some were experimental prompts, and one was an external example of how a hidden carrier can be projected into a visible relation. None of those roles is mathematical proof evidence for Eqs. (4), (8), or (11).

The current version preserves that history without exposing private chat bodies or local-drive paths. It also preserves the unit-distance analogy: projection arguments must define the carrier, visible relation, projection map, multiplicity control, and theorem boundary. The analogy helped organize proof questions, but it is not a dependency of either Lean file.

This provenance separation is lossless in the sense that every valid v1 role remains represented:

Version 1 contribution	Version 2 disposition
source-route recovery	preserved as historical provenance
Jackie / Scott comparison context	preserved without private quotation
unit-distance proof archetype	preserved as methodology
condensate / clock / resonance leads	preserved as experimental routing
finite proof obligations	replaced by the later formal result
higher theorem cautions	restated as an explicit scope ladder

9. What This Adds to the SIT Program

The finite Farey theorem contributes three reusable operations to the broader Super Information Theory research program.

First, it supplies a precise finite carrier on which every object and relation can be enumerated. Second, it demonstrates that a failed classification should not be confused with failure of the target. Third, it provides a machine-checkable constructor route that can be cited at its exact strength.

These operations matter for any later projection or coherence theorem. A larger SIT result would still need to specify:

1. the physical or analytic carrier corresponding to the finite arithmetic model;
2. the projection or readout map;
3. the conditions preventing collapse or uncontrolled multiplicity;
4. the finite-to-infinite transfer;
5. the operator and spectral identification; and
6. the empirical interpretation.

The current theorem closes none of those mappings automatically. Its positive scope is Eq. (4) in the finite formalization. Finite C-22, an infinite weighted operator theorem, a zeta-zero spectral identification, the Coherence Tower, HSIT, an SIT / Riemann bridge, the Riemann Hypothesis, and physical SIT remain separate targets with their own definitions and proof obligations.

That boundary is not a weakness of the finite result. It is what makes the result reusable: later work can depend on a clearly typed theorem rather than on an ambiguous global proof label.

10. Discriminating Tests and Failure Conditions

The final-preprint package is designed to fail visibly if its claims are overstated or its proof objects drift.

10.1 Source-integrity test

Every promoted Lean file has a SHA-256 identity. Any edit changes the hash and invalidates the replay receipt until the file is checked again.

10.2 Placeholder and axiom test

The validator rejects `sorry`, `admit`, locally declared axioms, or `unsafe` tokens in promoted proof sources. It also records an axiom-dependency report for each terminal theorem, so native-decision or new logical dependencies cannot be silently hidden.

10.3 Definition test

The independent enumerator implements the public definitions rather than calling the Lean functions. A mismatch between the paper and formal source can therefore produce a divergent census.

10.4 Route-separation test

The package must preserve both facts:

- the historical endpoint-shape bridge is false as stated; and
- the direct constructor proves the finite listed-mediator target.

Deleting either result would distort the proof history.

10.5 Scope test

A valid citation to this paper must identify the finite carrier and terminal theorem. A citation that treats the result as an infinite spectral theorem, an RH proof, or physical validation fails the scope test.

11. Conclusion

The result is both mathematical and methodological. Mathematically, the replayed Lean development proves that determinant-greater-than-one endpoint pairs in the stated finite Farey carrier possess a listed strict mediator. Methodologically, it records why a failed endpoint taxonomy did not defeat the target and how Euclidean descent supplied a direct replacement route.

The final-preprint correction also strengthens the proof’s trust boundary. The historical native-decision steps were replaced by kernel reduction, their axiom reports were rechecked, the BC-055M reduction was verified inside the terminal source, and more than one million finite target cases were independently enumerated without failure.

This is the appropriate level at which to reuse the result: a source-identified, Lean-checked finite theorem with a preserved obstruction, a direct constructor, and explicit interfaces for whatever larger theory may later build upon it.

Data and Code Availability

The release package accompanying this preprint contains:

- the corrected BC-054IS Lean source;
- the BC-055O Lean source;
- the pinned Lean / Lake / Mathlib environment records;
- theorem and axiom reports;
- the independent finite enumerator and generated result;
- source and PDF checksums; and
- a machine-readable validation receipt.

Private source conversations are not required to reproduce the theorem and are not included.

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